

CSE106 Discrete Mathematics

Credit Hours: $3 + 0 = 3$

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Textbook:

Kenneth H. Rosen, *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*, 7th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2015.[5th and 6th Edition will also be sufficient]

Logic and Proofs

Propositional Logic

Proposition:

A proposition is a *declarative sentence* that is either **True** or **False**, but not both.

Example Following declarative sentences are propositions

1. Dhaka is the capital of Bangladesh (True)
2. Rajshahi is the capital of Bangladesh (False)
3. $1 + 1 = 2$ (True)
4. $2 + 2 = 3$ (False)

Example: Following sentences are not propositions

1. What time is it? (Not a declarative sentence)
2. Read this carefully. (Not a declarative sentence)
3. $x + 1 = 2$ (A declarative sentence but the truth value can be determined when the value of x is known)
4. $x + y = z$ (A declarative sentence but the truth value can be determined when the values of x , y , and z are known)

Propositional variables:

Propositions are represented using variables. The conventional letters used as propositional variables are p, q, r, s , etc.

Truth value of a proposition:

Truth value of a proposition is True (T) if it is a true proposition and False (F) if it is a false proposition.

Compound propositions:

Compound propositions are formed from existing propositions using **logical operators**.

Definition 1:

Negation or NOT operator ($\neg$): Changes the truth value of the proposition.

Example:

p : Today is Friday. (False)

$\neg p$: Today is **not** Friday. (True)

Example:

p : At least 10 inches of rain fell today in Dhaka. (False)

$\neg p$: Less than 10 inches of rain fell today in Dhaka. (True)

Table: The Truth Table for the Negation (NOT) of a Proposition.

p	$\neg p$
T	F
F	T

Connectives:

The logical operators that are used to form new propositions from two existing propositions are called connectives.

Definition 2:

Conjunction or AND operator ($\wedge$): The conjunction $p \wedge q$ is True when both p and q are true and False otherwise.

Example:

p : Today is Friday.

q : It is raining today.

$p \wedge q$: Today is Friday **and** it is raining today.

It is True on Friday when it is raining (rainy Friday).

It is False on days other than Friday.

It is False on days that are not raining.

It is False on Fridays when it is not raining.

Table: The Truth Table for the Conjunction (AND) of Two Propositions.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Definition 3:

Disjunction or OR operator ($\vee$): The disjunction $p \vee q$ is False when both p and q are False and True otherwise.

Example:

p : Today is Friday.

q : It is raining today.

$p \vee q$: Today is Friday **or** it is raining today.

It is True on Friday.

It is True on all rainy days.

It is True on Friday when it is raining.

It is False on days other than Friday when it is not raining.

Table: The Truth Table for the Disjunction (OR) of Two Propositions.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Definition 4:

Exclusive OR operator ($\oplus$): The *exclusive or* $p \oplus q$ is True when exactly one of p and q is True and is False otherwise.

Example:

p : Today is Friday.

q : It is raining today.

$p \oplus q$: **Either** today is Friday **or** it is raining today.

It is False on Friday when it is raining.

It is True on Friday when it is not raining.

It is True on days other than Friday when it is raining.

It is False on days other than Friday when it is not raining.

Table: The Truth Table for the Exclusive OR of Two Propositions.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Conditional Statements

Definition 5:

Implication or conditional operator ($\rightarrow$): The implication or conditional statement $p \rightarrow q$ is the proposition “if p , then q .” It is False when p is true and q is false, and True otherwise.

p is called the *hypothesis* (or *antecedent* or *premise*).

q is called the *conclusion* (or *consequence*)

Example:

p : Matin learns CSE.

q : Matin will find a good job.

$p \rightarrow q$: **If** Matin learns CSE, **then** he will find a good job.

It is True if Matin learns CSE and finds a good job.

It is False if Matin learns CSE and does not find a good job.

It is True if Matin does not learn CSE and finds a good job.

It is True if Matin does not learn CSE and does not find a good job.

Table: The Truth Table for the Implication $p \rightarrow q$.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

CONVERSE, CONTRAPOSITIVE, AND INVERSE

Implication: $p \rightarrow q$

Converse of $p \rightarrow q$: $q \rightarrow p$

Contrapositive of $p \rightarrow q$: $\neg q \rightarrow \neg p$

Inverse of $p \rightarrow q$: $\neg p \rightarrow \neg q$

Equivalent Compound Propositions:

When two compound propositions always have the same truth value, then they are **equivalent**.

An implication and its contrapositive are equivalent, that is, $p \rightarrow q \equiv \neg q \rightarrow \neg p$.

p	q	$\neg q$	$\neg p$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

The converse and the inverse of an implication are equivalent, that is, $q \rightarrow p \equiv \neg p \rightarrow \neg q$.

p	q	$\neg p$	$\neg q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$
T	T	F	F	T	T
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	T	T	T

Example:

What are the contrapositive, the converse, and the inverse of the following conditional statement?

The home team wins whenever it is raining.

Solution:

The original statement can be rewritten as

If it is raining, then the home team wins. ($p \rightarrow q$)

The contrapositive is

If the home team does not win, then it is not raining. ($\neg q \rightarrow \neg p$)

The converse is

If the home team wins, then it is raining. ($q \rightarrow p$)

The inverse is

If it is not raining, then the home team does not win. ($\neg p \rightarrow \neg q$)

BI-CONDITIONALS

Definition 6:

Bi-implication or bi-conditional or equivalence operator ($\leftrightarrow$): The bi-implication or bi-conditional or equivalence statement $p \leftrightarrow q$ is the proposition “if and only if p , then q ”. It is True when p and q have the same truth values, and is False otherwise.

Table: The Truth Table for the Bi-implication $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The bi-implication “if and only if p , then q ” is equivalent to “if p , then q and if q , then p ”.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Example:

p : You can take the flight.

q : You buy a ticket.

$p \leftrightarrow q$: You can take the flight if and only if you buy a ticket.

If you took the flight, then you bought a ticket and if you bought a ticket, then you took the flight.

Truth Tables of Compound propositions**Example:**

Construct the truth table of the compound proposition $(p \vee \neg q) \rightarrow (p \wedge q)$.

Table: The Truth Table of $(p \vee \neg q) \rightarrow (p \wedge q)$

$p \ q$	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T T	F	T	T	T
T F	T	T	F	F
F T	F	F	F	T
F F	T	T	F	F

Precedence of Logical Operations

Table: Precedence of Logical Operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$\neg p \wedge q$ means $(\neg p) \wedge q$

$p \wedge q \vee r$ means $(p \wedge q) \vee r$

$p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

Translating English Sentences

Example:

You can access the Internet from campus only if you are a computer science major or you are not a freshman.

Let

a : You can access the Internet from campus.

c : You are a computer science major.

f : You are freshman.

The sentence can be represented as

$$a \rightarrow (c \vee \neg f)$$

Example:

You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.

Let

q : You can ride the roller coaster.

r : You are under 4 feet tall.

s : You are older than 16 years old.

The sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q$$

Propositional Equivalences

Definition 1:

A compound proposition that is always true is called a *tautology*.

A compound proposition that is always false is called a *contradiction*.

A proposition that is neither a tautology nor a contradiction is called a *contingency*.

Example:

Table: Examples of a Tautology and a Contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

$p \vee \neg p$ is a tautology

$p \wedge \neg p$ is a contradiction

Logical Equivalences

Compound propositions that have the same truth values in all possible cases are called **logically equivalent**.

The notation $p \equiv q$ denotes that p and q are logically equivalent.

Table: De Morgan's Laws.

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

De Morgan's laws extend to

$$\neg (p_1 \vee p_2 \vee \dots \vee p_n) \equiv (\neg p_1 \wedge \neg p_2 \wedge \dots \wedge \neg p_n)$$

$$\neg (p_1 \wedge p_2 \wedge \dots \wedge p_n) \equiv (\neg p_1 \vee \neg p_2 \vee \dots \vee \neg p_n)$$

Example: Show that $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Table: Truth Tables for showing $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

$p \ q$	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T T	T	F	F	F	F
T F	T	F	F	T	F
F T	T	F	T	F	F
F F	F	T	T	T	T

From the truth table we see that $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Example: Show that $p \rightarrow q \equiv \neg p \vee q$.

Table: Truth Tables for showing $p \rightarrow q \equiv \neg p \vee q$.

$p \ q$	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T T	F	T	T
T F	F	F	F
F T	T	T	T
F F	T	T	T

From the truth table we see that $p \rightarrow q \equiv \neg p \vee q$.

Example: Show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ [*Distributive law*].

Table: A Demonstration that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

$p \ q \ r$	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T T T	T	T	T	T	T
T T F	F	T	T	T	T
T F T	F	T	T	T	T
T F F	F	T	T	T	T
F T T	T	T	T	T	T
F T F	F	F	T	F	F
F F T	F	F	F	T	F
F F F	F	F	F	F	F

From the truth table we see that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.